Dadmehr Daniel Ghasemfar

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Technical Skills

Languages: Java, Python, C++, C, MATLAB, Maple, TensorFlow, HTML/CSS/JavaScript, ROS/ROS2
Software: CAD (SolidWorks, Fusion, OnShape), FEA and CFD (Ansys, SolidWorks, In-House), PCB (Eagle, Altium), LabVIEW, Gazebo, Git
Equipment: 3D Printing (SDF, SLS, Metal), Machining (Milling, CNC, Lathe, Woodwork), Lasercutting, Soldering
Theoretical: PDE's and ODE's, Numerical Analysis, Accurate GPU Simulations, PID and Advanced Controls, Structural Analysis, Thermal Analysis, Transfer Functions, Vibration Analysis, Systems Integration
Soft Skills: Project Lead, IP Strategy, Investor Negotiations, Fluent Farsi, Conversational Spanish

Education

Duke University — Durham, NC
Bachelor of Science in Mechanical Engineering with Minors in Math and Psychology — Aug 2020 – Dec 2024

- Engineering GPA: 3.6 → Math GPA: 3.5
- Classes: intermediate robotics, intermediate dynamics, chaos and perturbation theory, controls, heat and mass transfer, fluid dynamics, algorithms, data structures, ODE/PDE's, numerical analysis, solids and statics, aircraft performance and design
- Graduated as a Pratt Fellow for recognition of a 2+ year research project.

Experience

Atar Energy Solutions [thinktank] — August 2024 – Present
Co-Founder — Alexandria, VA

- Designed and prototyped an integrated high-voltage tester, achieving a 67% reduction in cost and weight versus market competitors, securing a competitive edge in a $700M sector.
- Led business strategy by negotiating with investors and IP-law firms, resulting in a finalized patent plan and legal entity formation to protect company intellectual property.
- Collaborated on a 120-page technical datasheet and a 70-page professional plan.

Duke General Robotics Lab — Aug 2022 – May 2024
Individual Research — Durham, NC

- Worked under Dr. Boyuan Chen to independently build and develop a versatile and accurate parallel-processed wave propagation simulation. This could emulate any material and lattice, any 3D shape, and input forces. Used to generate artificial data to train a custom A.I. that can "see with sound".
- Developed a novel validation method using Chladni plate analysis, bridging simulation and experimental data with 90% accuracy to train an AI model for acoustic perception.
- Research approved by Duke Engineering Board for *Graduation with Distinction award.*

FPrin LLC — Oct 2020 – Feb 2021
Lab Technician — Mountain View, CA

- Designed, built, and operated case-specific test devices to effectively characterize a recently patented German compressed gas-powered insulin pen. After that, stress tested thousands of components and maintained the shop/lab.
- Engineered custom test fixtures for a patented medical device, enabling the successful characterization and stress-testing of thousands of components to ensure product reliability and performance.

Edsall Auto Services — January 2025 – Present
Car Mechanic - Part Time — Alexandria, VA

- Perform mechanical repairs and diagnostics on 30+ vehicle models, demonstrating hands-on troubleshooting skills.

Projects

- Led a 6-person team in the design and testing of an electro-resistive thruster, achieving 162s ISP and 1.36N thrust to meet client requirements aligned with future NASA grants.
- Designed and fabricated a 2-foot-tall, 8-DOF bipedal robot; developed a Python-based control system for dynamic movement, resolving complex hardware integration issues like mechanical instability and solder joint failures.
- Awarded a Mayor's Medal for developing a CNN in TensorFlow, achieving 94% classification accuracy for forest fire detection in drone video feeds.
- FRC Robotics Quarter-finalist
- 2x Synopsys Science Fair Finalist (Engineering and CS)